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QUESTIONS

1. In an approximation, the difference between the exact solution (x_{true}) and the approximate solution ($x_{approximate}$) is called the *true absolute error* and its formula is: $E_t = |x_{true} - x_{approximate}|$. However, you have learnt a second type of error: *relative error* ($E_r = \left|\frac{x_{true} - x_{approximate}}{x_{true}}\right|$). Why might we need to use relative error? Please explain with a simple example that we use relative error in numerical calculations to show that the approximation is more meaningful/better. (10 pts)

2. Let the function $f(x)$ be continuous on the interval $[a, b]$. How can you guarantee that the equation $f(x) = 0$ has at least one real root in the interval $[a, b]$? Explain briefly. (10 pts)

3. Let $f(x) = \frac{e^x}{1+x^2}$ and $x_0 = 0$. Determine the Taylor expansion about x_0 with the first three terms. Calculate the approximate value of the $f(x)$ function by taking $x = 0.125$ in the expansion you obtained. (15 pts)

4. Consider the equation $x - cos(x) = 0$.
   a. Prove that this equation has at least one real root in the interval $[1, 2]$. (5p).
   b. Calculate the approximate root of the given equation within the interval $[1, 2]$ using the Regula Falsi method after two iterations. (10p)

5. Find the root of the equation $\sqrt{x} + x^2 = 7$ using Newton's method. Start at $x = 7$ and carry out the first five iterations. (20 pts)

6. Consider this function: $f(x) = x^3 - x - e^x - 2$

   An approximate solution will be calculated using the fixed-point iteration method with the initial condition $x_0 = 2$.
   a. For this function; find the iteration formula using the initial condition. (Note: This function has four different iteration formula candidates. Please write and explain all.) (20p)
   b. Using this iteration formula, calculate the approximate root after 2 iterations. (10p)

2. Let the function $f(x)$ be continuous on the interval $[a, b]$. How can you guarantee that the equation $f(x) = 0$ has at least one real root in the interval $[a, b]$? Explain briefly. (10 pts)

if $f(a).f(b) < 0$ so there is at least one solution root

3. Let $f(x) = \frac{e^x}{1+x^2}$ and $x_0 = 0$. Determine the Taylor expansion about x_0 with the first three terms. Calculate the approximate value of the $f(x)$ function by taking $x = 0.125$ in the expansion you obtained. (15 pts)

$$f(x) \approx f(x_0) + f'(x_0).(x-x_0) + \frac{f''(x_0).(x-x_0)^2}{2}$$

$$\frac{e^0}{1} + \frac{e^0.1 - e^0.0}{1}.x + \frac{(e^0.-1.1) - e^0.1.2.0}{1.2}.x^2$$

$$1 + 1.x + \frac{(-1-0).x^2}{2}$$

$$= 1 + x - \frac{x^2}{2}$$

$x = 0.125$

$$= 1.1171$$

$$f'(x) = \frac{e^x.(1+x^2) - e^x.(2x)}{(1+x^2)^2} \quad \begin{matrix} \to u(x) \\ \to v(x) \end{matrix}$$

$e^x(1+x^2-2x) = e^x.(x-1)^2$

$$f''(x) = \frac{u'(x).v(x) - u(x).v'(x)}{(v(x))^2}$$

$$= \quad u(x) = e^x.(x-1)^2$$
$$u'(x) = e^x.(x-1)^2 + e^x.2.(x-1)$$

$$v(x) = (1+x^2)^2$$
$$v'(x) = 2.(1+x^2).2x$$

$$f''(x) = \frac{e^x\left((x-1)^2 + 2(x-1)\right)\cdot(1+x^2)^2 - e^x\cdot(x-1)^2\cdot 2\cdot(1+x^2)\cdot 2x}{(1+x^2)^4}$$

4. Consider the equation $x - \cos(x) = 0$.

a. Prove that this equation has at least one real root in the interval [1,2]. (5p).

b. Calculate the approximate root of the given equation within the interval [1,2] using the Regula Falsi method after two iterations. (10p)

0,5

changed

[0.5, 1]

a) $f(1) = 1 - \cos(1) = 0.4596$

$f(0.5) = 0.5 - \cos(0.5) = -0.3775$

$>$ $f(1)\cdot f(0.5) < 0$ so at least one real root in the interval

b) regula-falsi

$$\frac{a\cdot f(b) - b\cdot f(a)}{f(b) - f(a)} =$$

1st iterat.

$i = 1$

$$x_1 = \frac{0.5\cdot(0.4596) - 1\cdot(-0.3775)}{0.4596 - (-0.3775)} = \frac{0.2298 + 0.3775}{0.8371}$$

$x_1 = 0.7254$

$f(x_1) \rightarrow f(0.7254) = -0.0228$ $\quad f(1) \cdot f(0.7254) < 0$ so new interval is $[0.7254, 1]$

2st iterat. $\quad i = 2$ $\quad x_2 = \dfrac{(0.7254).(0.4596) - 1.(-0.0228)}{0.4596 - (-0.0228)} = \dfrac{0.3333 + 0.0228}{0.6876} = 0.5178 \rightarrow x_2$

5. Find the root of the equation $\sqrt{x} + x^2 = 7$ using Newton's method. Start at $x = 7$ and carry out the first five iterations. (20 pts)

$f(x) = \sqrt{x} + x^2 - 7 = 0$

$f'(x) = \frac{1}{2} \cdot x^{-1/2} + 2x$

Newton method

$x_{i+1} = x_i - \dfrac{f(x_i)}{f'(x_i)}$

1st iterat. $\quad i = 0$ $\quad x_1 = x_0 - \dfrac{f(x_0)}{f'(x_0)} = 7 - \dfrac{44.645}{14.188} = 3.85$

2st iterat. $\quad i = 1$ $\quad x_2 = x_1 - \dfrac{f(x_1)}{f'(x_1)} = 3.85 - \dfrac{9.784}{7.954} = 2.619$

3st iterat.

$i = 2$

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)} = 2.619 - \frac{f(2.619)}{f'(2.619)} \rightarrow 2.619 - \frac{\overbrace{1.477}^{0.2737}}{5.395} = 2.3653$$

6. Consider this function: $f(x) = x^3 - x - e^x - 2$

An approximate solution will be calculated using the fixed-point iteration method with the initial condition $x_0 = 2$.

a. For this function; find the iteration formula using the initial condition. (Note: This function has four different iteration formula candidates. Please write and explain all.) (20p)

b. Using this iteration formula, calculate the approximate root after 2 iterations. (10p)

1st cand.

$$x^3 = x + e^x + 2$$
$$x = \sqrt[3]{x + e^x + 2}$$
$$g(x) = \sqrt[3]{x + e^x + 2}$$
$$g'(x) = \frac{1}{3} \cdot (x + e^x + 2)^{-2/3} \cdot (1 + e^x)$$
$$g'(2) = \frac{1}{3} \cdot (4 + e^2)^{-2/3} \cdot (1 + e^2)$$

$= 0.552 < 1$ so we can use this formula

2nd candidate

$$x = x^3 - e^x - 2$$
$$g(x) = x^3 - e^x - 2$$
$$g'(x) = 3x^2 - e^x$$
$$g'(2) = 4.61$$

$g'(2) \not< 1$ we can't use this formula

3rd candidate

$$e^x = x^3 - x - 2$$
$$x = \ln(x^3 - x - 2)$$
$$g(x) = \ln(x^3 - x - 2)$$
$$g'(x) = \frac{3x^2 - 1}{x^3 - x - 2}$$
$$g'(2) = \frac{11}{4} \not< 1$$

so we can't use this formula

4th candidate

$$x^3 - x - e^x - 2 = 0$$
$$x(x^2-1) - e^x - 2 = 0$$

$$x.(x^2-1) = e^x + 2$$
$$x = \frac{e^x+2}{x^2-1}$$

$$g(x) = \frac{e^x+2}{x^2-1}$$

$$g'(x) = \frac{e^x.(x^2-1) - (e^x+2).(2x)}{(x^2-1)^2}$$

$$g'(2) = \frac{e^2.3 - (e^2+2).4}{9} = -1.70$$

$g'(2) < 1$? so we can't use this formula

iteration formula

$x_0 = 2$

$$x_{i+1} = \sqrt[3]{x_i + e^{x_i} + 2}$$

$i=0$ $\quad x_1 = \sqrt[3]{2+e^2+2} = 2.249$

$i=1$ $\quad x_2 = \sqrt[3]{2.249 + e^{2.249} + 2.} = 2.3943$